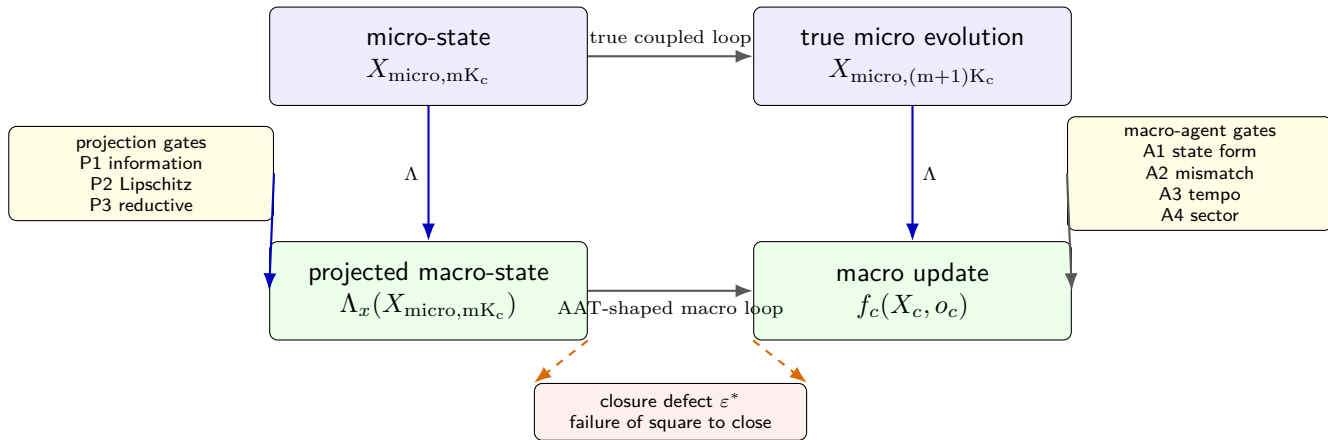


bridge lemma: under incremental contraction,
per-step defect becomes bounded trajectory error



audit watch: scalar gate needs route typing, norm weighting, and information-set discipline